FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE JEE (Advanced), 2015

FULL TEST – V

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 234**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Section-A (13 to 18) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

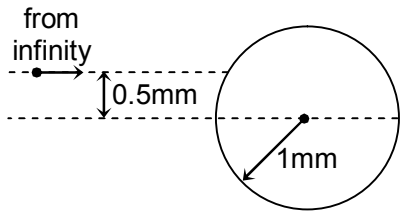
PART – I

SECTION – A

Straight Objective Type

This section contains **8 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. A particle of mass 1 kg and charge $\frac{1}{3}\mu C$ is projected towards a non-conducting fixed spherical shell having same charge uniformly distributed on its surface. Find minimum initial velocity of projection required by the particle, if it just grazes the shell :



(A) $\sqrt{\frac{2}{3}}$ m / s (B) $2\sqrt{\frac{2}{3}}$ m / s

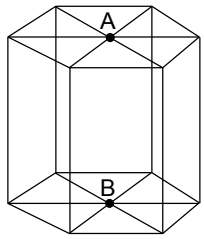
(C) $\frac{2}{3}$ m / s (D) None of these

2. High-Quality camera lenses are coated to prevent reflection. A lens has an optical index of refraction 1.72 and a coating with an optical index of refraction of 1.31. For near-normal incidence, the minimum thickness of the coating to prevent reflection for wavelength of 5.3×10^{-7} m is:

(A) 0.75 μm (B) 0.2 μm

(C) 0.1 μm (D) 1.75 μm

3. In the diagram shown, all the straight segments have resistance 'R'. The equivalent resistance between the upper dot (A) and lower dot (B) is

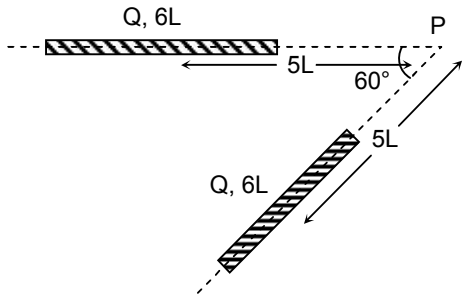


(A) $\frac{3R}{8}$ (B) R

(C) $\frac{R}{2}$ (D) $\frac{2R}{5}$

Space for Rough work

4. Two rods each of length 6L carry charge Q are placed as shown in figure. Find magnitude of electric field at point P, which is at a distance of 5L from centre of each rod.



(A) $\frac{1}{64\pi\in_0}.\frac{\sqrt{3}Q}{L^2}$

(B) $\frac{\sqrt{5}Q}{4\pi\in_0 L^2}$

(C) $\frac{1}{2\pi\in_0}.\frac{Q}{L^2}$

(D) $\frac{3Q}{5\pi\in_0 L^2}$

5. A particle moves along x-axis following the relation $px^2+qv^2=r$, where p, q, r are positive constants. x is co-ordinate of the particle and v is the instantaneous speed. Find the time interval between two consecutive instants, when particle is at rest.

(A) $2\pi\sqrt{\frac{q}{p}}$ (B) $\pi\sqrt{\frac{q}{p}}$

(C) $\pi\sqrt{\frac{p}{q}}$ (D) $2\pi\sqrt{\frac{p}{q}}$

6. A charged particle of specific charge $\left(\frac{\text{charge}}{\text{mass}}\right)$ 'α' is released from origin at t = 0 with velocity $\vec{v}=v_0(\hat{i}+\hat{j})$ in uniform magnetic field $\vec{B}=B_0\hat{i}$. Then co-ordinates of the particle at $t=\frac{\pi}{B_0\alpha}$ will be :

(A) $\left(\frac{v_0}{2B_0\alpha},\frac{\sqrt{2}v_0}{\alpha B_0},-\frac{v_0}{B_0\alpha}\right)$ (B) $\left(-\frac{v_0}{2B\alpha},0,0\right)$

(C) $\left(0,\frac{2v_0}{B_0\alpha},\frac{v_0\pi}{2B_0\alpha}\right)$ (D) $\left(\frac{v_0\pi}{B_0\alpha},0,-\frac{2v_0}{B_0\alpha}\right)$

Space for Rough work

7. A resistor 'R' is connected across a source of emf 'E' and unknown internal resistance. Power of resistor is maximum when internal resistance is
(A) R (B) < R
(C) > R (D) zero

8. A child loves to watch as you fill a transparent plastic bottle with shampoo. Horizontal cross-sections of the bottle are circles with varying diameters. You pour in shampoo with constant volume flow rate 16.5 cm^3/s. At what rate is its level in the bottle rising at a point where diameter of the bottle is 6.30 cm. Calculate keeping significant figures in view.
(A) 0.063 cm/s (B) 0.63/cm/s
(C) 0.529 cm/s (D) 0.5291 cm/s

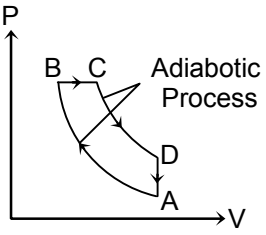
Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. A boy in the elevator with open roof shoots a bullet in vertical upward direction from a height of 1.5 above the floor of the elevator. The initial speed of the bullet with respect to elevator is 15 m/s. The bullet strikes the floor after 2 seconds. Then (Assuming g = 10 m/s^2)
(A) Lift is moving with constant speed
(B) Lift is moving with upward acceleration of 5.75 m/s^2
(C) Lift is moving with downward acceleration of 5.75 m/s^2
(D) Distance travelled by bullet during its flight can not be calculated from the given data.

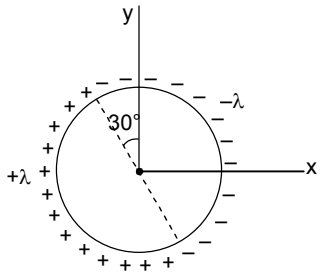
Space for Rough work

10. An idealized diesel engine operates in a cycle known as air standard diesel cycle as shown. Fuel is sprayed into the cylinder at the point of maximum compression, B. Combustion occurs during expansion BC.



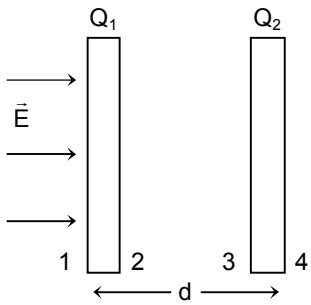
(A) Heat is absorbed during process BC.
(B) Heat is released during process DA.
(C) Heat is absorbed during process CD.
(D) Efficiency of the cycle is $\left[1-\frac{1}{\gamma}\left(\frac{T_D-T_A}{T_C-T_B}\right)\right]$

11. Consider a non-conducting ring of radius 'a' in x-y plane. The two semi-circular portions have linear charge densities + λ and – λ as shown. Then :



(A) Dipole moment of the ring is $-\left(2\sqrt{3}\,\hat{i}+2\hat{j}\right)\lambda a^2$
(B) Dipole moment of the ring is $-\left(2\hat{i}+2\sqrt{3}\,\hat{j}\right)\lambda a^2$
(C) Electric field at the center of the ring is $\frac{\lambda}{2\pi\epsilon_0 a}\left(\sqrt{3}\,\hat{i}+\hat{j}\right)$
(D) Electric field at the center of the ring is $\frac{\lambda}{2\pi\,\epsilon_0\,a}\left(\hat{i}+\sqrt{3}\,\hat{j}\right)$

12. Two plates, each of one side area 'A' are kept at a separation 'd'. They have charge Q_1 and Q_2 and uniform electric field $\vec{E}$ exist in the region as shown. Then



(A) Charge on surface (1) is $\frac{Q_1+Q_2}{2}-A\,\epsilon_0\,E$
(B) Field inside the gap is $E+\frac{Q_1-Q_2}{2A\,\epsilon_0}$
(C) Field outside the gap is E
(D) Field on either side of the plates is identical

Space for Rough work

Comprehension Type

This section contains **3 paragraphs**, each describing theory, experiments, data etc. Six questions relate to the three paragraphs with **two questions on each paragraph**. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Questions 13 & 14

Two astronauts, each having a mass 'M' are connected by a rope of length 'd' having negligible mass. They are isolated in space, orbiting their center of mass at speed 'v'. Treat astronauts as particles. By pulling the rope, one astronaut shortens the distance between them to '$\frac{d}{2}$'.

13. Calculate work done by the astronaut in shortening the rope :
    (A) $2Mv^2$ (B) $3Mv^2$
    (C) Mv^2 (D) zero

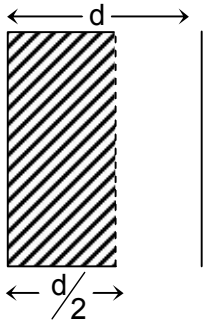
14. Physical quantity that will remain conserved during the process of pulling in rope.
    (A) Speed of astronauts
    (B) Energy of the system
    (C) Angular speed of astronauts about COM
    (D) Angular Momentum

Space for Rough work

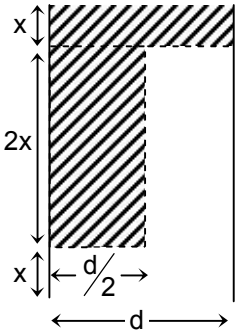
Paragraph for Questions 15 & 16

A parallel plate capacitor has plate area 'A' and plate separation 'd'. A dielectric medium of volume $\frac{Ad}{2}$ and dielectric constant 'K' is available. To use it in the capacitor, two alternative methods are available as shown :

(1)



(2)



15. Capacitance in case (1) is $X_1\frac{\in_0 A}{d}$. Then x_1 is

(A) $\frac{K}{3}$ (B) $\frac{2K}{k-1}$

(C) $\frac{3K+1}{2K}$ (D) $\frac{2K}{K+1}$

16. Capacitance is case (2) is $x_2\frac{\in_0 A}{d}$. Then x_2 is

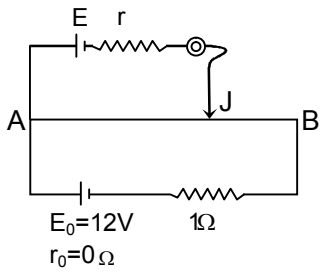
(A) $\frac{K+1}{4}+\frac{K}{K+1}$ (B) $\frac{K+1}{4}+\frac{1}{K}$

(C) $\frac{2K}{4}+\frac{K}{3}$ (D) $\frac{K}{3}+\frac{K+1}{2}$

Space for Rough work

Paragraph for Questions 17 & 18

In the set-up shown 'E' is unknown emf with internal resistance 'r'. AB is a uniform potentiometer wire, 11m long and resistance $1\Omega m^{-1}$. When AJ = 1m, current through galvanometer is 2.4 A when AJ = 10m, current through galvanometer is $\frac{6}{11}$A, in the opposite direction as in previous case. Galvanometer has negligible resistance.



17. The value of E is
(A) 4 V (B) 6 V
(C) 8 V (D) 10 V

18. The vale of r is
(A) 1Ω (B) 2Ω
(C) 2.5Ω (D) 4Ω

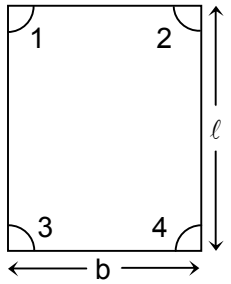
Space for Rough work

SECTION – C

Integer Answer Type

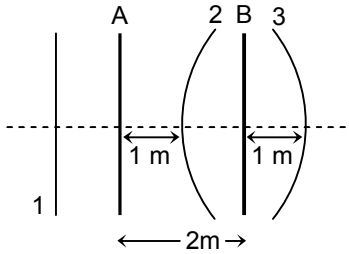
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Ball A is at rest somewhere on a smooth billiards table $\left(\frac{\ell}{b} = 3\right)$. An identical ball B moving with velocity 2 ms^{-1}, collides with a elastically. After $\frac{1}{2}$ sec both the balls simultaneously fall into pockets marked 1 and 2. Find area of the table in m^2.



2. A hydrogen atom in excited state n reaches the ground state by emitting three photons of frequency in ratio 5 : 72 : 243. Wavelength corresponding to one of the transitions belongs to visible region. Find the value of 'n'.

3. A and B are thin lenses. 1, 2 and 3 are representative wave fronts of incident and refracted monochromatic light with distances from the lenses as shown. The radii of curvature of wave fronts 2 and 3 are 4m and 7m respectively. Find the focal length (modulus value) of lens B in meter.



4. A particle is taken to a height K times radius of Earth, above the surface of earth and projected at an angle of 30° to the vertical. During its subsequent motion, it just grazes the surface of earth with a velocity $\frac{\sqrt{15}}{4}$ times the escape velocity from the surface of earth. Find the value of K. (K < 9).

5. ABCD is a semi-circular wire carrying current in a region of uniform magnetic field. Arc AB = Arc BC = Arc CD. The magnitude of the force on the wire is 8 N, while magnitude of the force on arc AB alone is 9 N. The lines of action of these two forces include an angle of $\cos^{-1}\left(\frac{2}{3}\right)$. Find the magnitude of the force (is N) on arc CD alone.

Space for Rough work

Chemistry PART – II

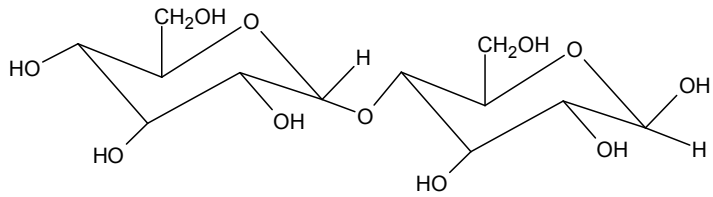
SECTION – A

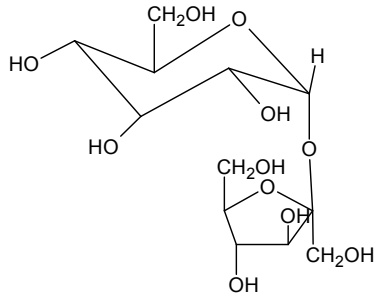
Straight Objective Type

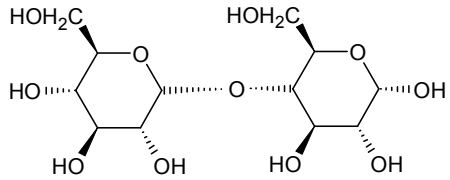
This section contains **08 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** is/are correct.

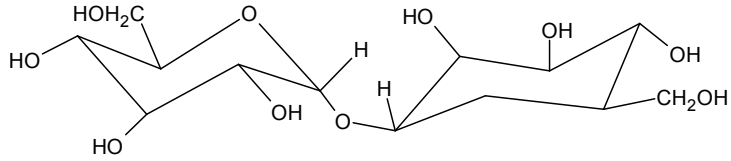
1. In a rock salt crystal and fluorite structure, if atoms along the plane cutting the two opposite edges are removed, then the new ratio of the cations in two respective unit cells and ratio of anions in respective unit cells are:
   (A) 5 : 6 / 10 : 1 (B) 6 : 5 / 5 : 8
   (C) 12 : 5 / 5 : 8 (D) 3 : 4 / 5 : 4

2. Which of the following is reducing sugar?

(A)


(B)


(C)


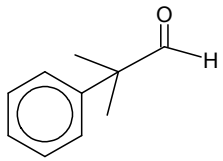
(D)


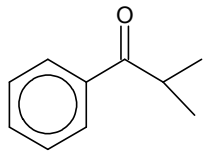
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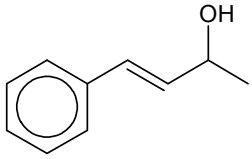
3. In compound (A) with molecular formula C_8H_8O, which responds to iodoform test?

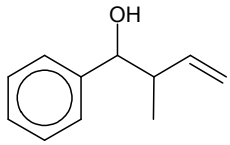
$$(A) \xrightarrow[HOEt]{NaCN} (B) \xrightarrow[(ii)\ H^+, H_2O]{(i)\ MeMgBr} (C) \xrightarrow{H^+} \underset{\text{(Major product)}}{(D)}$$

The structure of (D) is:

(A) 

(B) 

(C) 

(D) 

4. When copper is reacted with dil. HNO_3 a colourless gas is liberated. Which of the following is true for the gas?
(A) Has 11 valence e's, where extra electron occupies an antibonding π^*2p orbital.
(B) An acid anhydride.
(C) Paramagnetic in solid state.
(D) Has no resonance.

5. What is the number of five membered ring in the molecule of S_4N_4?
(A) 4 (B) 2
(C) 1 (D) 5

6. Consider the following complexes:
(I) $[Cr(CO)_x)]$ (II) $[Cr(CO)_{x-1}PF_3]$
If PF_3 is better π accepter than CO, what will be the order of bond length of CO in complexes (I) and (II):
(A) I > II (B) II > I
(C) I = II (D) cannot be compared

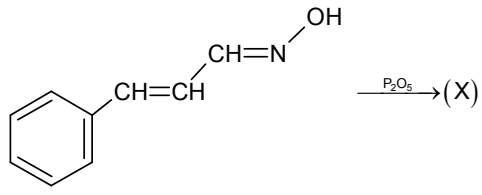
7. A compound (A) form a unstable pale blue colour solution in water. The solution decolourised Br_2 water and an acidified solution of $KMnO_4$. The possible compound (A) is:
(A) HNO_2 (B) HNO_3
(C) N_2O_3 (D) Both (A) and (C)

8. The 100 ml of 0.1 M CH_3COOH ($K_a = 10^{-5}$) is titrated with 0.1 M NaOH and pH is observed at three stages, at 298 K.
1st step $\Rightarrow$ 50 ml of 0.1 M NaOH added.
2nd step $\Rightarrow$ 100 ml of 0.1 M NaOH added.
3rd step $\Rightarrow$ 120 ml of 0.1 M NaOH added.
Calculate the difference in pH between stages (I & II) and (II & III) stages.
(A) $(pH_{II} - pH_I) = 3.84$ & $(pH_{III} - pH_{II}) = 4.11$ (B) $(pH_I - pH_{II}) = 2$ & $(pH_{III} - pH_{II}) = 5.2$
(C) $(pH_I - pH_{II}) = 3.84$ & $(pH_{III} - pH_{II}) = 3.11$ (D) $(pH_I - pH_{II}) = 4.2$ & $(pH_{III} - pH_{II}) = 3.11$

Space for Rough work

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. 

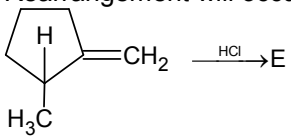
(A)

Which of the following are correct for compound (X)?
(A) (X) is aromatic and a heterocyclic compound.
(B) Compound (X) is more basic than aniline.
(C) The reaction that occur is stereospecific.
(D) The reaction do not follow a concerted mechanism and (X) is non-aromatic.

10. Which of the following the first pair has less parameter in the property mentioned:

(A) CH_3NH_2 or $(CH_3)_2NH_2^+$	(Angle H – N – H)
(B) PCl_2F_3 or PCl_3F_2	(Angle F – P – F)
(C) NO^+ or NO	(Bond length)
(D) CrO_3 or Cr_2O_3	(In acidic nature)

11. Which of the following statements are incorrect?
(A) 1, 1, 1-trideutero – 2-propanol reacts with conc. H_2SO_4 at high temperature to give only one alkene 3,3,3- trideuteropropene.
(B) Attack of chlorine radical on alkene is endothermic and attack of radical formed on HCl is exothermic.
(C) Rearrangement will occur in forming the major product in reaction.



(D) *Cis* form of 1,3-Dimethyl cyclohexane is more stable than its *trans* form with respect to atleast one group at equatorial position.

12. During discharging of lead storage battery, which of the following is/are true?

(A) H_2SO_4 is produced.	(B) H_2O is consumed.
(C) $PbSO_4$ is formed at both electrodes.	(D) Density of electrolytic solution increased.

Space for Rough work

Comprehension Type

This section contains **3 paragraphs**, each describing theory, experiments, data etc. Six questions relate to the three paragraphs with **two questions on each paragraph**. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Question Nos. 13 to 14

A substance 'X' dissociates by two parallel reactions with different Arrhenius constant (A_1 & A_2) and different activation energies $\left(E_{a_1} \& E_{a_2}\right)$ as shown:

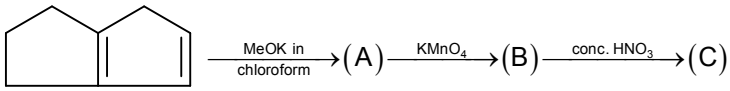
$$X(g) \xrightarrow[E_{a_1}=12\ \text{kcal/mol}]{A_1=2\times10^{-14}\ s^{-1}} Y(g);\ X(g) \xrightarrow[E_{a_2}=9\ \text{kcal/mol.}]{A_2=2\times10^{-14}\times e^{-2}s^{-1}} Z(g)$$

13. Temperature at which net activation energy for dissociation of X will be 10.5 kcal/mol?
(A) 200 (B) 750
(C) 800 (D) 1000

14. Which of the following statements are correct with respect to major product obtained in the above parallel decay?
(A) 'Y' will be the major product at temperature 300 K.
(B) 'Z' will be the major product at temperature 800 K.
(C) Both statements are correct.
(D) None of the statement are correct.

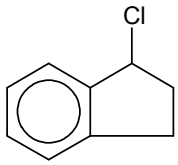
Space for Rough work

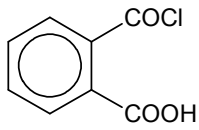
Paragraph for Question Nos. 15 to 16



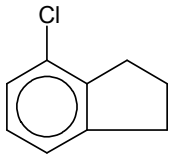
15. Identify the structure of (A) & (B):

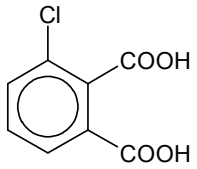
(A)

A ⇒ 

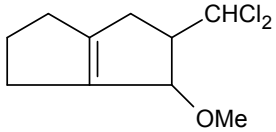
B ⇒ 

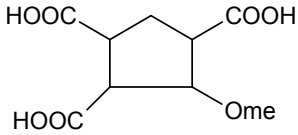
(B)

A ⇒ 

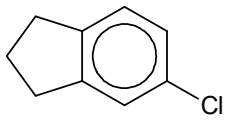
B ⇒ 

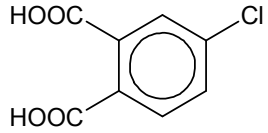
(C)

A ⇒ 

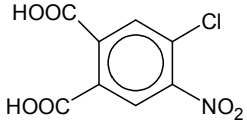
B ⇒ 

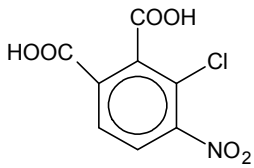
(D)

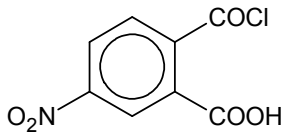
A ⇒ 

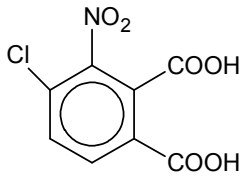
B ⇒ 

16. How many isomers of (C) are possible and the major product is:

(A) 3, 

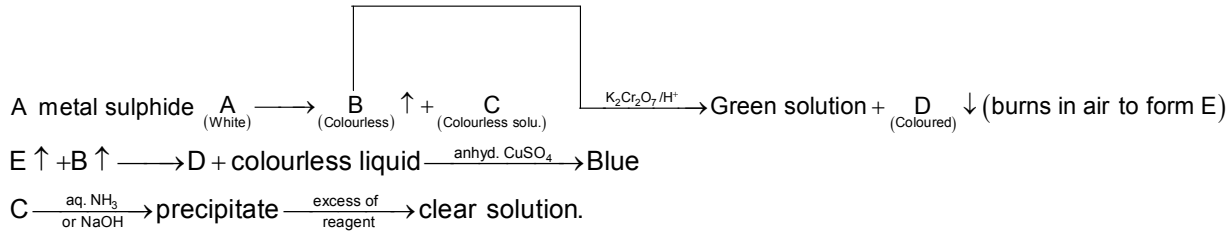
(B) 3, 

(C) 4, 

(D) 2, 

Space for Rough work

Paragraph for Question Nos. 17 to 18



17. Calculate the weight of compound (A) required to produce (B), so that (B) will completely react with 10 ml of 0.2 M $K_2Cr_2O_7$?
(A) 650 mg (B) 288 mg
(C) 583 mg (D) 458 mg

18. Compound (C) on reaction with NH_3 or NaOH forms a precipitate and with excess of reagent forms a clear solution of compound (Z), (Z) is?
(A) $NaAlO_2$ (B) Na_2ZnO_2
(C) $Cu(OH)_2$ (D) $[Ag(NH_3)_2]^+$

Space for Rough work

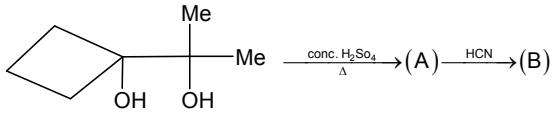
SECTION – C

Integer Answer Type

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. How many are the intensive properties?
   Density, pressure, emf, free energy, specific heat, heat capacity, specific conductance, resistance, pH?

2. The sum of chiral centre and number of lone pair in compound (B) are:



3. One mole of an ideal monoatomic gas is expanded irreversibly in two stage isothermal expansion:
   Stage 1. (8 bar, 4.0 litre, 300 K)
   Stage 2. (2 bar, V_1 litre, 300 K)
   Stage 3. (1 bar, V_2 litre, 300 K)
   Calculate heat absorbed by the gas in the process (in kJ)

4. The sum of all sp^3, sp^2 & sp units present in solid PCl_5, solid Pbr_5 and solid N_2O_5 are:

5. How many diastereomeric pair are possible for compound $[Co(en)_2Cl_2]^+$?

Space for Rough work

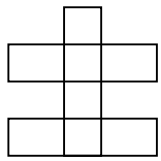
Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. Six different letters are placed in boxes of the given figure such that no row is empty. (In each box only one letter can be placed). The number of such arrangements is



(A) 2920 (B) 6480
(C) 8640 (D) 10800

2. Suppose b and c are real numbers randomly chosen in the interval [0, 1]. What is the probability that the distance in complex plane between the two roots of the equation $z^2 + bz + c = 0$ is not greater than 1?

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$
(C) $\frac{1}{4}$ (D) $\frac{3}{4}$

3. The equation of line which intersect each of two lines $2x + y - 1 = 0 = x - 2y + 3z$ and $3x - y + z + 2 = 0$, $4x + 5y - 2z - 3 = 0$ and is parallel to $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ is

(A) $4x + 7y - 6z - 3 = 0$, $2x - 7y + 4z + 7 = 0$ (B) $4x + 7y - 6z - 4 = 0$, $2x - 7y + 4z + 2 = 0$
(C) $4x + 7y - 6z - 2 = 0$, $2x - 7y + 4z + 5 = 0$ (D) none of these

4. If three positive distinct real number p, q, r satisfy the condition $\frac{1}{p} + \frac{1}{r} + \frac{1}{p-q} + \frac{1}{r-q} = 0$. If q = 2, then minimum value of pr is

(A) 1 (B) 2
(C) 3 (D) 4

Space for rough work

5. The value of $\int_0^{\infty} \frac{1}{\left(1+x^{17}\right)\left(1+x^2\right)} dx$ is equal to

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

6. If a curve is such that line joining origin to any point P(x, y) on the curve and line parallel to y-axis through P are equally inclined to tangent to curve at P, then the differential equation of the curve is

(A) $x\left(\frac{dy}{dx}\right)^2 - 2y\frac{dy}{dx} = x$ (B) $\left(\frac{dy}{dx}\right)^2 + 2y\frac{dy}{dx} = x$

(C) $y\left(\frac{dy}{dx}\right)^2 + 2y\frac{dy}{dx} = x$ (D) $y\left(\frac{dy}{dx}\right)^2 - 2y\frac{dy}{dx} = y$

7. Let f (x) = $x^3 + 2x^2 - x + 1$, then which of the following statement(s) is/are correct

(A) $\sin^{-1}\left(\sin\left([\alpha]\right)\right) + \cos^{-1}\left(\cos\left([\alpha]\right)\right) = 6 - \pi$ (where α is real root of f(x) = 0 and [.] denotes the G.I.F.)

(B) f (x) = 0 has three distinct real roots

(C) y = f (x) is increasing function

(D) $\sin^{-1}\left(\sin\left([\alpha]\right)\right) + \cos^{-1}\left(\cos\left([\alpha]\right)\right) = 5 - 2\pi$

(where α is real root of f(x) = 0 and [.] denotes the G.I.F.)

8. Which of the following functions is an injective (one–one) function in their respective domain?

(A) f(x) = 2x + sin 3x (B) f(x) = x · [x], (where [.] denotes the G.I.F)

(C) f(x) = $\frac{2^x - 1}{4^x + 1}$ (D) f(x) = $\frac{2^x + 1}{4^x - 1}$

Space for rough work

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct

9. Let point A be (1, 1) and equation of BC is $(x + y - 1) + \lambda(2x - y + 4) = 0$ and area of ΔABC is maximum, then slope of the line AB and AC, may be equal to
(Given AB = AC and BC = 2 units)

(A) $\frac{-2-\sqrt{5}}{2\sqrt{5}-1}$ (B) $\frac{2-\sqrt{5}}{2\sqrt{5}-1}$

(C) $\frac{2-\sqrt{5}}{2\sqrt{5}+1}$ (D) $\frac{2+\sqrt{5}}{2\sqrt{5}+1}$

10. The value of x which satisfy the equation $\begin{vmatrix} \tan x & \tan 2x & \tan 3x \\ \tan 2x & \tan 3x & \tan x \\ \tan 3x & \tan x & \tan 2x \end{vmatrix} = 0$ is

(A) $\frac{13\pi}{12}$ (B) $\frac{13\pi}{3}$

(C) $\frac{\pi}{3}$ (D) none of these

11. The co-ordinate of foci of ellipse $\frac{(x-y)^2}{4} + \frac{(y+x)^2}{1} = 1$ are

(A) $\left(\sqrt{\frac{3}{2}}, -\sqrt{\frac{3}{2}}\right)$ (B) $\left(-\sqrt{\frac{3}{2}}, -\sqrt{\frac{3}{2}}\right)$

(C) $\left(\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}}\right)$ (D) $\left(-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}}\right)$

12. For different values of k, the circle $x^2 + y^2 + (8 + k)x + (8 + k)y + (16 + 12k) = 0$, always passes through two fixed points P and Q. For $k = k_1$, then tangents at P and Q intersect at the origin. Which of the following is/are correct?

(A) the mid-point of P and Q is (–6, –6) (B) the sum of ordinates of P and Q is –12

(C) k_1 may be equal to $-\frac{32}{9}$ (D) k_1 may be equal to $\frac{8}{3}$

Space for rough work

Comprehension Type

This section contains **3 paragraphs**, each describing theory, experiments, data etc. Six questions relate to the three paragraphs with **two questions on each paragraph**. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Question Nos. 13 to 14

Let continuous function f (x) = $\begin{cases} x^2 + 10x + 8, & x \le -2 \\ px^2 + qx + r, & -2 < x < 0 \\ x^2 + 2x, & x \ge 0 \end{cases}$, where $p \ne 0$. If a line 'L' touches the graph of y = f (x) at 3 points, then

13. The line 'L' touches which of the following curve
(A) $x^2 + y^2 = 1$ (B) $x^2 + y^2 - 2x - 4y - 9 = 0$
(C) $(y + 2)^2 = -48 (x + 1)$ (D) $(y + 2)^2 = 13 (x + 1)$

14. The value of p + q + r is equal to
(A) $5\sqrt{2}$ (B) 5
(C) 6 (D) 7

Paragraph for Question Nos. 15 to 16

Read the following write up carefully and answer the following questions:
Consider some special type of matrices. A square matrix is called idempotent matrix if $A^2 = A$
A matrix A is called nilpotent matrix if $A^k = 0$, for some $k \in N$
A square matrix is called involutary matrix if $A^2 = I$
Now consider the following matrices $A = \begin{vmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{vmatrix}$, $B = \begin{vmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{vmatrix}$, $C = \begin{vmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{vmatrix}$

15. Which one of the following is a nilpotent matrix?
(A) A (B) B
(C) C (D) AC^2

16. Which one of the following is not an idempotent matrix?
(A) A^3C^2 (B) A^2C^2
(C) BC^2 (D) C^2A

Space for rough work

Paragraph for Question Nos. 17 to 18

Read the following write up carefully and answer the following questions:
$x^2 + y^2 - 2ax - 2y - 8 = 0$, where a is variable.

17. The equation represents a family of circle passing through two fixed point whose coordinates are
(A) (0, –2), (0, 2) (B) (0, –2), (0, 4)
(C) (4, 0), (2, 0) (D) none of these

18. Equation of a circle C of this family, tangent to which at these fixed points intersects on the line $2x + y + 5 = 0$ is
(A) $x^2 + y^2 - 2x - 8y - 8 = 0$ (B) $x^2 + y^2 - 2x + 6y - 8 = 0$
(C) $x^2 + y^2 - 2x + 8y - 8 = 0$ (D) $x^2 + y^2 - 6x - 2y - 8 = 0$

SECTION – C

Integer Answer Type

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. n balls are arranged to form an equilateral triangle such that first row consists of one ball, second row consists of two balls, third row consists of three balls and so on. 49 more balls are added to form a square. If each side of the square contains 3 balls more than that of the equilateral triangle, then sum of digits of n is _____

2. Let O, $A(z_1)$, $B(z_2)$, $C(z_3)$ are vertices of rhombus such that $|z_1| = |z_2| = 4$ and $|z_3| = 6$ if $\arg(z_2 - z_1) = \frac{\pi}{3}$ then $\arg(z_3)$ is $\frac{3\pi}{k}$ then k is equal to _____

3. If $\int_0^{\pi} \frac{\cos x}{(x+4)^2} dx = P$ and $I = \int_0^{\pi/2} \frac{\sin 2x}{\pi + 4 - 2x} dx$, then the value of $P + 2I - \frac{1}{\pi + 4}$ is equal to $\frac{\lambda}{12}$ then λ is _____

4. The value of $\tan 1^\circ \tan 2^\circ + \tan 2^\circ \tan 3^\circ + ... + \tan 88^\circ \tan 89^\circ$ is equal to $\cot^2 1^\circ - n$, where n is a two digit number ab, then the value of b – a is _____

5. Let $\frac{1}{a_1 - 2i}, \frac{1}{a_2 - 2i}, \frac{1}{a_3 - 2i}, \frac{1}{a_4 - 2i}, \frac{1}{a_5 - 2i}, \frac{1}{a_6 - 2i}, \frac{1}{a_7 - 2i}, \frac{1}{a_8 - 2i}$ are vertices of regular octagon. If the area of octagon is A, then the value of $8\sqrt{2}A$ is
(where $a_i \in R$ for i = 1, 2, 3, 4, 5, 6, 7, 8) _____

Space for rough work